**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU – UGP (HONOURS)**

**FIRST SEMESTER EXAMINATION**

(2024 ADMISSION ONWARDS)

**MG1MDCECT102 Data Analytics**

**Duration: 1 Hour** **Max.Marks:35**

*Students should attempt at least one question from each course outcome to enhance their overall outcome attainability*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C)*

Multiple Choice Questions Answer **35** Questions
Each question carries **one** mark

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| 1 | What is the primary purpose of data visualization in analytics?<br>a) To increase data storage b) To make data easier to understand<br>c) To reduce the amount of data d) To generate random data | U | 2 |
| 2 | Which type of visualization is best suited for showing trends over time?<br>a)Pie chart b) Histogram c) Line chart d) Box plot | K | 2 |
| 3 | What type of data is most suitable for visualizing using a line chart?<br>a) Nominal data b) Ordinal data<br>c) Continuous data d) Categorical data | U | 2 |
| 4 | Which is the best practice for labeling the lines in a line chart?<br>a) Use generic names b) Use abbreviations<br>c) Label each line clearly with a legend d) Do not label them at all | U | 2 |
| 5 | Which of the following is NOT a tool commonly used for diagnostic analytics?<br>a)SQL b) A/B testing c) Decision trees d) Regression analysis | U | 1 |
| 6 | What is the primary goal of diagnostic data analytics?<br>a) To predict future outcomes b) To understand why something happened<br>c) To optimize decision-making processes d) To collect raw data | U | 1 |
| 7 | What is the difference between "Save" versus "Save As"?<br>a) "Save" creates a backup copy<br>b) "Save As" allows you to rename the file or change its format<br>c) "Save" always prompts for a location<br>d) There is no difference; they are the same | U | 1 |
| 8 | In the "Save As" dialog, which option allows you to change the file's format?<br>a)Save b) File type dropdown c) Filename box d) None of the above | A | 1 |
| 9 | Which of the following formats cannot be imported into Excel?<br>a).xml b) .jpg c) .txt d) .xlsx | U | 1 |
| 10 | What is the primary function of the "Export" option in Excel?<br>a) To delete a file b) To save a workbook in a different format or location<br>c) To print a workbook d) To share a workbook via email | A | 1 |
| 11 | What is the primary purpose of data analytics?<br>a) Storing data b) Predicting future trends<br>c) Visualizing data only d) Collecting data | U | 1 |
| 12 | Which of the following tools is commonly used for data visualization in data analytics?<br>a) Python b) Excel c) Tableau d) Microsoft word | K | 1 |

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| 13 | Which sector has significantly benefited from data analytics in recent years<br>a) Only healthcare b) Only retail<br>c) All sectors, including healthcare, finance, and marketing<br>d) Only manufacturing | K | 1 |
| 14 | Which of the following tools is commonly used in computerized data analytics?<br>a)Microsoft Word b) Google Chrome c) SQL d) Adobe Photoshop | K | 1 |
| 15 | What type of data can computerized data analytics process?<br>a) Only structured data b) Only unstructured data<br>c) Both structured and unstructured data d) Only text-based data | U | 1 |
| 16 | Identify the three main components of the "3 Vs" of big data<br>a) Volume, Velocity, Variety b) Value, Variability, Visualization<br>c) Volume, Variability, Validity d) Velocity, Value, Visualization | K | 1 |
| 17 | Recognize a method used for analyzing big data.<br>a) Manual sorting b) Predictive analytics<br>c) Hand calculations d) Simple averaging | U | 1 |
| 18 | Recognize a source from where raw data is commonly collected<br>a) Reports b) Surveys c) Data warehouses d) Dashboards | K | 1 |
| 19 | Identify the role of raw data in data analytics<br>a) It is the final result b) It serves as the initial input for analysis<br>c) It is used only for data storage d) It replaces processed data | U | 1 |
| 20 | Identify a tool commonly used for storing and managing structured data<br>a) Microsoft Excel b) Adobe Photoshop<br>c) PowerPoint d) Notepad | K | 1 |
| 21 | Name the structure that organizes data in a relational database<br>a) Files b) Tables c) Images d) Directories | K | 1 |
| 22 | Name the graphical representation commonly used for discrete data<br>a) Histogram b) Line graph c) Bar chart d) Scatter plot | K | 1 |
| 23 | Recognize a discrete type of variable<br>a) The number of books in a library b) The speed of a car<br>c) The time spent running d) The volume of water in a bottle | U | 1 |
| 24 | Name the type of data that represent continuous data<br>a) Discrete values b) Countable values<br>c) Measurable values that can take any value within a range d) Categorical values | K | 1 |
| 25 | Name the graphical representation commonly used for continuous data<br>a) Pie chart b) Bar chart c) Line graph d) Scatter plot | K | 1 |
| 26 | Select the best method for collecting continuous data<br>a) By counting distinct items in a set<br>b) By measuring with a device like a ruler or thermometer<br>c) By asking yes/no questions<br>d) By grouping data into categories | U | 1 |
| 27 | Name a secondary data source<br>a) Surveys conducted by the researcher<br>b) Online databases of research papers<br>c) Observations made during an experiment<br>d) Data collected through personal interviews | K | 1 |
| 28 | Name a common source of secondary data for businesses<br>a) Customer interviews b) Published industry reports<br>c) Focus group discussions d) In-house surveys | K | 1 |
| 29 | What is qualitative data? | U | 1 |

a) Data that is numerical and can be measured.
b) Data that is always continuous.
c) Data collected through scientific instruments.
d) Data that describes characteristics or categories.

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| 30 | What is the main purpose of using quantitative data?<br>a) To explore meanings and experiences<br>b) To describe personal opinions<br>c) To measure and compare quantities<br>d) To analyze behaviours in a descriptive, non-numerical way | U | 1 |
| 31 | Which of the following is numerical data type?<br>a) Customer feedback b) Age of individuals<br>c) Favourite food d) Gender | U | 1 |
| 32 | Which of the following is a type of navigation data commonly used in GPS systems?<br>a) Satellite signals b) Sales transactions<br>c) Weather forecasts d) Social media metrics | U | 1 |
| 33 | Which of the following can be a source for importing data into Excel?<br>a) CSV files b) Web pages c) Databases d) All of the above | U | 2 |
| 34 | What is the main purpose of conditional formatting in Excel?<br>a) Change font style b) Highlight data<br>c) Change colour d) Import data | A | 2 |
| 35 | The result of the formula = MIN(1,2,3,4) is<br>a)1 b) 2 c) 3 d) 4 | A | 2 |
| 36 | What is a survey in the context of data collection?<br>a) A method of observing behavior in real-time<br>b) A tool for collecting large amounts of data via questions<br>c) A technique for conducting experiments<br>d) A process for collecting secondary data | U | 2 |
| 37 | Which of these is a common survey question format?<br>a) Coding tasks b) Multiple-choice questions<br>c) Experimental observations d) Mathematical equations | U | 2 |
| 38 | Which of the following is a qualitative data collection method?<br>a) Survey b) Interview c) Questionnaire d) Data mining | U | 2 |
| 39 | Which of the following is a common method of collecting data in observational research?<br>a) Surveys b) Interviews c) Video recording d) Data mining | U | 2 |
| 40 | What does the term "outlier" refer to in data analysis?<br>a) A duplicate data entry<br>b) A missing value in a dataset<br>c) A data point that significantly differs from the majority of the data<br>d) The mean of the dataset | U | 2 |

**1* 35 = 35**